

JAMES CHIBUIKE OKORIE

Embedded Systems Laboratory, Department of Physics, University of Cape Coast

Cape Coast, CR, Ghana * (+233) 543-456-605 * james.okorie@stu.ucc.edu.gh * okoriejames84@gmail.com

LinkedIn profile: <https://www.linkedin.com/in/james-c-okorie>

PERSONAL PROFILE

Applied Physicist and Embedded Systems Specialist with expertise at the intersection of Edge AI, IoT, and precision hardware prototyping. Proven track record in conducting field-ready research, from solar-powered agricultural monitoring platforms to accessible biomedical devices, alongside extensive university-level teaching and lab demonstration experience. Adept at translating complex physical concepts into practical technology solutions while fostering student learning and collaborative research.

EDUCATION

- Expected 11/2026 *University of Cape Coast: Department of Physics*
MPhil. Physics
Relevant Courses: Data Acquisition Systems, Advanced Quantum Mechanics, Computing and Numerical Methods, Quantum Statistics, Mathematical Methods, Microprocessor I and II, Graduate Seminar, Academic Writing for Graduate Studies.

Thesis: Solar-Powered Thermal Imaging System for Real-Time Detection and Monitoring of Fall Armyworm Infestation in Maize Farms: A Precision Agriculture Approach for Smallholder Farmers in Benin.
- 08/2018 – 11/2022 *University of Cape Coast*
BSc. Engineering Physics
Thesis: RFID-Based Examination Attendance System
Honors: Second Class honors (Upper division) GPA: 3.4/4.0

RESEARCH EXPERIENCE

- 02/2025 – 02/2026 Research Scholar, Intra-African PATH Project
*Genetics, Biotechnology & Seed Sciences Laboratory (GBioS),
University of Abomey-Calavi, Abomey-Calavi, Benin*
- Developing a solar-powered thermal imaging system for the early detection of Fall Armyworm in maize as part of a research team.
 - Engaged in advanced research on intelligent agricultural systems, contributing to climate-smart solutions.
- 11/2022 – 02/2025 Laboratory & Research Assistant
*Embedded Systems Laboratory, Department of Physics
University of Cape Coast, Cape Coast, Ghana*
- Contributed to departmental research projects, including the design and implementation of a traffic light control system, while collaborating with lecturers on planning and project discussions.
 - Supervised undergraduate research projects and organized practical sessions, ensuring adherence to laboratory procedures and guiding students in experimental design and technical reporting.

RESEARCH PROJECTS

Applied Research and Development Projects

- Estimation of Nitrogen N, Phosphorus P, and Potassium in the Soil Using Regression Analysis
- Arduino-Based Orange Fruit Defect Detection using Resistance and Two-Point Probe Measurement
- Development of an Arduino-Based System for Dynamic Fluid Volume Determination
- Monitoring of Soil moisture content using a resistive moisture sensor via Arduino
- Smart Sight: A smart home application targeted toward low-vision patients
- Arduino-Based Automated Diaper Monitoring Device
- A study of the design and construction of a Low-cost egg Hatching System
- Simulation and Testing of an Arduino-Based Titration System

SKILLS

Technical Skills & Tools

- **Programming:** Python, C++, RStudio, MATLAB, Simulink, Embedded Development: Arduino IDE, VS Code (PlatformIO)
- **AI & ML:** Edge AI, OpenCV, TinyML.
- **Robotics & Hardware:** Arduino (ESP32, Nano RP2040, Nano BLE Sense 33, Uno, Mega 2560), Raspberry Pi, ROS 1 & ROS 2 (self-learned), Circuit Design, PCB Prototyping.
- **Software:** QGIS, KiCAD, Proteus, FreeCAD, Overleaf LaTeX, Git/GitHub.

Research & Analytical Skills

- Scientific Research Methodologies, Experimental Design & Implementation, Data Acquisition & Interpretation, Problem-Solving & Critical Thinking, Literature Review & Synthesis, Technical Report Writing, Statistical Analysis.

Soft Skills & Professional Attributes

- Teamwork & Collaboration, Leadership & Mentorship, Effective Communication (Written & Oral), Time Management & Organization, Adaptability & Fast Learning, Project Management, Presentation Skills.

PROFESSIONAL EXPERIENCE

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|-------------------|---|
| 01/2023 - Present | <p>Independent Research & Development Project</p> <ul style="list-style-type: none">▪ Actively involved in the design and development of innovative low-cost Transcranial Photobiomodulation (tPBM) brain wellness cap and a PBM knee brace, to support recovery and accessibility in healthcare.▪ Designed and produced 5+ prototypes across multiple wavelengths (630 nm, 860 nm, 1068nm), with 3 shipped for testing and use, demonstrating practical application and iterative engineering under the mentorship of Dr. Clifford S. Saunders. |
| 10/2024 - 02/2025 | <p>Demonstrator
<i>Department of Physics, University of Cape Coast</i>
Cape Coast, Ghana</p> <ul style="list-style-type: none">▪ Assisted undergraduate students during practical laboratory sessions and provided academic support for coursework.▪ Represented my supervisor in lectures and facilitated learning in their absence.▪ Contributed to the assessment process through invigilation, grading, and marking of quizzes and scripts. |
| 04/2025 - 07/2025 | <p>Data Science and Machine Learning Intern
<i>Edureka, Online</i></p> <ul style="list-style-type: none">▪ Developed hands-on expertise in data collection, preprocessing, and machine learning model building using Python and key Machine Learning libraries.▪ Applied data analysis, statistical modeling, and visualization techniques under mentorship, strengthening practical knowledge and problem-solving skills |

- 11/2022 - 10/2023 Teaching Assistant
Department of Physics, University of Cape Coast
Cape Coast, Ghana
- Assisted lecturers in teaching undergraduate students, organizing research data, and planning departmental research.
 - Designed assessment tools and provided tutoring/mentorship to enhance student learning and progress.

PUBLICATIONS

Kayil, B., **Okorie, J. C.**, Achigan-Dako, E. G., Fassinou Hotegni, N. V., Tchokponhoue, D. A., Houdegbe, A. C., Raeth, P., Quartermaine Bastin, G., Ezin, E. C., Dompheh, K. A. Damage, Detection and Management of Fall Armyworm (*Spodoptera frugiperda* J.E. Smith) in Benin's Maize Production Systems. (**In Review**)

CONFERENCES AND WORKSHOPS

Selected Conferences and Workshops

- 3rd African Conference on Precision Agriculture (AfCPA), Precision Agriculture in Action for Africa, University of Cape Coast, Ghana, Satellite Station **Satellite Panellist & Participant**
- Ghana-Norway Summer School in Medical Physics and Radiography, Emergence of New Technologies in Diagnostic Imaging, University of Cape Coast **16th - 20th October 2023, Participant**
- Low-cost Optical Spectroscopy Techniques for Vector (Mosquito) Surveillance, Laser and Fiber Optic Center (LAFOC), University of Cape Coast **21st - 25th August 2023, Participant**

VOLUNTEERING

- 10/2023 - Present Robotics Engineer Lead
Tech4Everyone, Cape Coast, Ghana
- Design and deliver robotics courses, create comprehensive tutorials, and guide students on STEM projects.
- 07/2024 - 02/2025 Tutor, Python Programming
Latter-Day Saints, Abura Stake YSA
Cape Coast, Ghana
- Delivered comprehensive lessons on Python programming fundamentals (syntax, control flow, data structures, project development) to foster practical coding skills.
 - Mentored students through hands-on exercises and collaborative projects, enhancing problem-solving abilities and practical application of coding concepts.
- 04/2019 Sanitation Committee Member
ITI-CCR Annual National Conference, Ghana
- Volunteered as a sanitation committee member to assist in keeping the campground and program environment neat for healthy living.

MEMBERSHIP

- 05/2025 - Present African Association of Precision Agriculture (AAPA)
08/2018 - Present Physics Student Association of Ghana (PHYSAG), UCC Chapter
08/2018 – 11/2022 National Society of Black Engineers (NSBE), UCC Chapter

REFERENCES

Available Upon Request